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Kinematic determinants of dive height in springboard diving

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Abstract. The aim of the present investigation was to identify the kinematic characteristics of dives that produce a greater peak flight height. Eight junior divers (13.25 ± 1.04 years) performed forward somersault dives at three different fulcrum settings off a one metre springboard. Selected variables were input to a stepwise regression analysis to identify predictors of dive height. Significant predictor variables were vertical velocity at touchdown, angle of lean at takeoff, vertical displacement at touchdown, ankle angle at touchdown, and external work. External work done during takeoff, although a significant predictor of dive height, was of relatively minor influence when compared to the kinematic variables. This study emphasises the importance of the hurdle flight in setting touchdown parameters that strongly determine take-off characteristics.

Key words: Springboard diving, height, kinematics, work

Résumé. Déterminants cinématique de la hauteur atteinte en phase aérienne lors du plongeon au tremplin.

L'objectif de cette étude était d'identifier les déterminants cinématiques du plongeon qui permettent de produire une plus grande élévation au cours de la phase de vol. Huit plongeurs juniors ($13,25 \pm 1,04$ ans) ont effectués au tremplin de un mètre des plongeon composés d'un salto avant suivant trois conditions de longueur de la planche. Les déterminants mécaniques de l'élévation ont été identifiés par une régression linéaire. Cette analyse a mis en évidence l'importance des variables suivantes : vitesse verticale au contact de la planche, angle d'inclinaison à l'éjection, déplacement vertical au contact de la planche, angle des chevilles au contact de la planche, et travail mécanique externe. Le travail mécanique externe à l'éjection, bien que prédictif de l'élévation, avait une influence relativement faible au regard des variables cinématiques. Cette étude a mis en évidence l'importance de la phase préparatoire dans l'ajustement de la phase de contact avec la planche qui détermine fortement les paramètres de l'éjection.

Mots clés : Plongeon au tremplin, élévation, cinématique, travail mécanique

1 Introduction

The sport of springboard diving requires athletes to propel themselves into the air off the springboard, perform movements of varying complexity in the air, and then enter the water with minimal splash. Forward running approach dives consist of at least three steps forward, a hurdle which is a higher final step, and a takeoff into the air (Fig. 1). The takeoff propels the diver up and away from the board, while providing the necessary momentum and body orientation for the performance of aerobatic movements. The height achieved during dive flight is of importance as it contributes directly to the score achieved by the diver (McCormick, Subbiah, & Arnold, 1982).

During depression of the board before takeoff, the springboard stores elastic energy which is subsequently released back to the diver as kinetic energy. The amount of energy stored and released from the board is determined by the mechanical energy possessed by the diver at impact;

and by the change in energy achieved by the diver during takeoff phase, largely resulting from extension of the legs adding to the amount of board compression.

The same model diving board (Maxiflex 'B', Duraflex International, Sparks, NV) is used in all major diving competitions and has been shown to behave as a linearly elastic spring (Sprigings, Stilling, Watson, & Dorotich, 1990). The stiffness of the board is adjustable by means of a moveable fulcrum with nine marked positions which can change the effective length of the board from 2.705 to 3.289 m (Miller, Osborne, & Jones, 1998). Rotating a wheel moves the fulcrum between these nine settings, with position nine being towards the rear of the board giving the largest effective board length in front of the fulcrum. Intermediate positions can also be selected if required.

Elite adult divers tend to use fulcrum positions towards the rear of the diving board, increasing the effective length and mass of the board and reducing its stiffness (Miller *et al.*, 1998). This looser board setting has been reported to result in a longer duration of the takeoff phase, increased

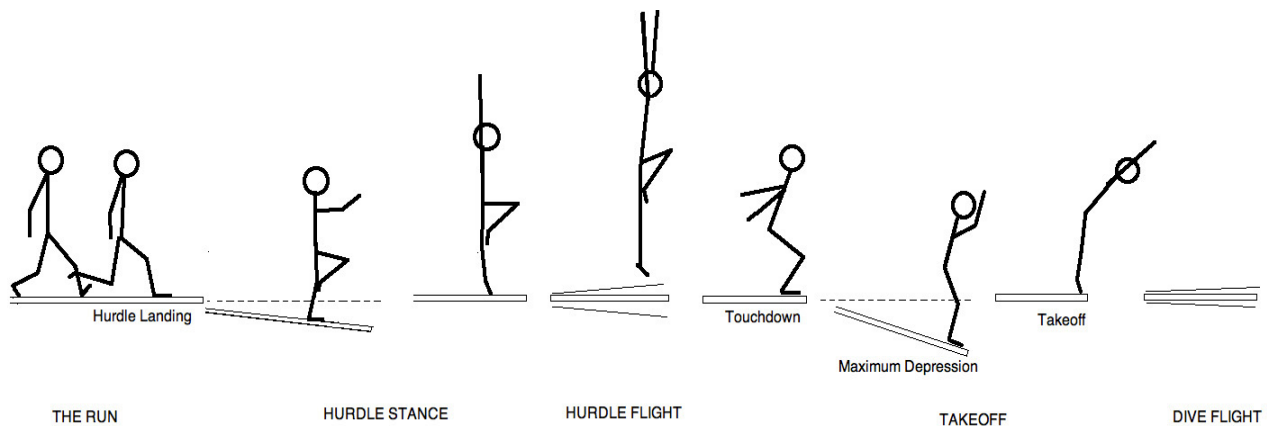


Fig. 1. Illustration of major phases in the springboard dive (Walker, 2010).

knee flexion at touchdown, increased maximum flexion of the knees and increased downward velocity of the board at touchdown (Jones & Miller, 1996). Jones and Miller (1996) also found that moving the fulcrum backward requires divers to have a higher hurdle flight to accommodate the lower frequency of board oscillation. Divers aim to land on the board after the hurdle flight at a time when the board has maximal downwards velocity. Landing at this time minimises impact velocity between the feet and the board and thus reduces the loss of energy at impact. When the fulcrum is positioned further back and the board oscillates more slowly, then a higher hurdle flight is required to increase flight time in order to land at the appropriate time in the board's oscillation cycle.

Younger divers tend to move the fulcrum further forward than elite divers, reducing the effective length and mass of the board and thus increasing its stiffness. This has been conjectured to be because a higher hurdle flight may be beyond their ability (Jones & Miller, 1996). Further, the increased amount of maximal knee flexion may require additional strength to compress the board. A final point, not mentioned by Jones and Miller, is that depressing the board further will generate a greater horizontal component to the force under the diver's feet, potentially increasing the difficulty of controlling movements during the dive.

Numerical simulations performed by Kooi and Kuipers (1994) suggested that a looser fulcrum setting would likely result in a higher dive flight; a finding supported by experimental measures by Sanders and Wilson (1988) with elite divers. It was unclear, however, whether high-level junior divers would similarly benefit from decreased board stiffness. Analysis by Walker (2010) demonstrated that there was no significant change in dive height with fulcrum setting (Fig. 2). There was, however, substantial variability between dives performed across all three fulcrum settings.

The present study represents a re-analysis of data collected by Walker (2010) to investigate the relationship between various kinematic measures and dive height.

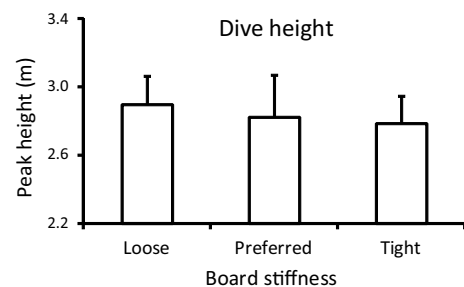


Fig. 2. Peak height resulting from dives at three different board stiffness settings (Walker, 2010).

Starting from the premise that fulcrum setting did not have a significant main effect on dive height, a regression analysis was performed to identify the variables which were most highly correlated with an increased dive height. The study aimed to identify those variables important to emphasise when coaching divers to achieve a higher dive flight.

2 Methods

Eight athletes were recruited from the New South Wales Institute of Sport (NSWIS) Silver and Emerging Talent Squads, following informed consent procedures approved by the University of Sydney human ethics committee. All athletes competed at a national level, representing their state at the Australian National Championships (Tab. 1). Each athlete reported that they were injury free and fit to perform the diving procedures.

Each subject performed five forward 1 $\frac{1}{2}$ piked somersault dives off a 1 m Maxiflex B springboard at three different fulcrum settings: their preferred setting (Tab. 1), minus one (tight) and plus one (loose) from the preferred. Each setting moves the fulcrum approximately 6.5 cm. The fulcrum was moved only one setting either side of the preferred position in order to avoid changing the feel of the board too much from that which the subjects normally use.

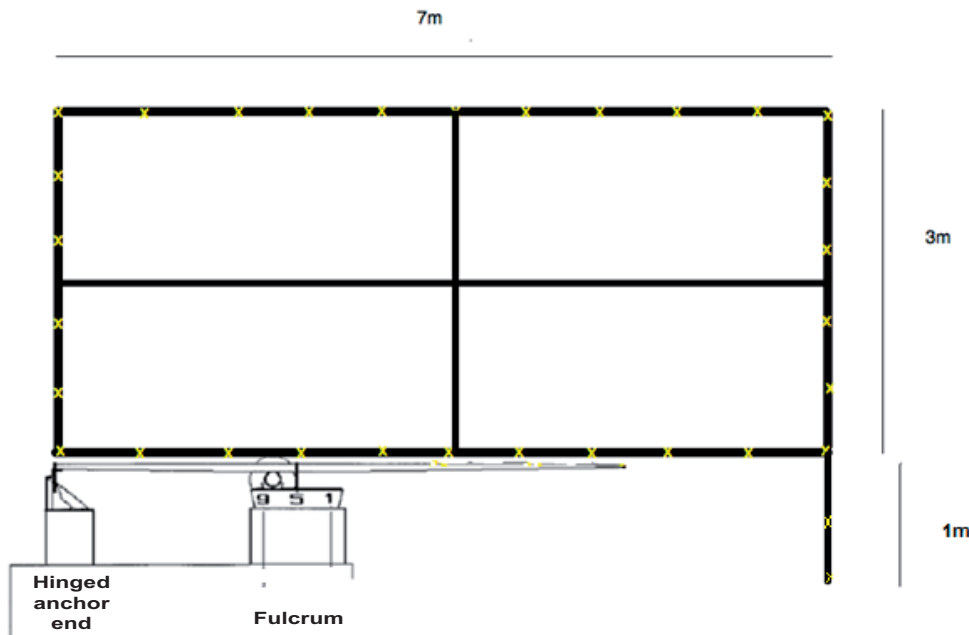


Fig. 3. Custom calibration frame (Walker, 2010).

Table 1. Subject characteristics. Mean $\pm$ SD, $n = 8$.

Age (years)	13.25 $\pm$ 1.04
Height (cm)	154.99 $\pm$ 9.44
Mass (kg)	44.32 $\pm$ 8.28
Training sessions/week	9.86 $\pm$ 0.83
Preferred fulcrum setting	6.25 $\pm$ 0.65

Prior to the experiment, subjects were given a 10 min warm up and time to familiarize with the springboard and environment. Subjects were then asked to perform two practice trials at each fulcrum setting before testing began. An accredited coach observed each dive to determine whether the trials were of a normal performance. Exclusion criteria included baulking of the dive, the buckling of the subject's legs and unsuccessful completion of the required dive.

Dives were filmed using a Casio Exilim EX-FH20 camera positioned perpendicular to and level with the springboard, on the hurdle support leg side of each subject. Prior to testing, a custom designed, 3 m $\times$ 7 m, frame was filmed for later calibration of digitised coordinates (Fig. 3). Non-toxic water proof paint was used to mark each subject's head of the 5th metatarsal, lateral malleolus of the fibular, lateral condyle of the femur, middle iliac crest, greater tubercle of the humerus, lateral epicondyle of the humerus and head of the ulnar on the side of the hurdle support leg (Fig. 4).

On a separate day, 95 anthropometric measurements were taken from the upper limb, lower limb, trunk and head in order to calculate individual inertial parameters for each subject using the methods of Yeadon (1990). The body was divided into seven segments: foot, shank, thigh, pelvis, trunk, upper arm and forearm. Yeadon's method

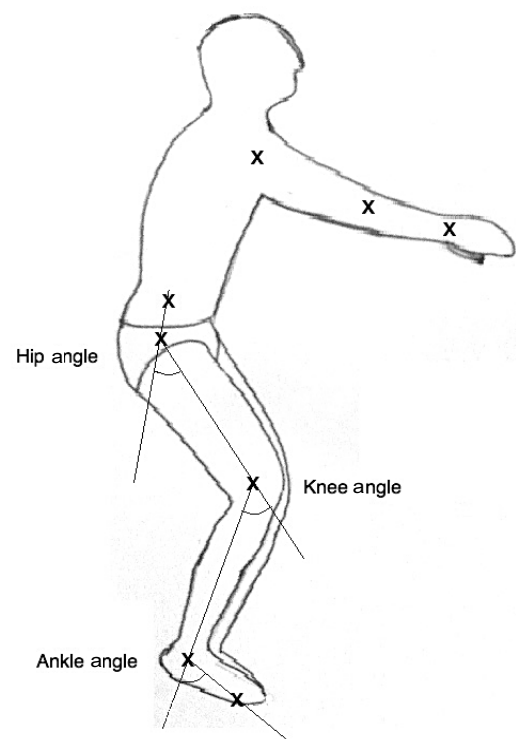


Fig. 4. Illustration of lower limb joint angles measured from video analysis (Walker, 2010).

estimated the mass and centre of gravity location for each of the seven segments.

The three best trials for each subject were digitised at a frame rate of 70 Hz using SkillSpector software (Video4Coach 2009). Displacement data were filtered

using a fourth order dual pass Butterworth filter. An analysis of residuals was used to select a filter frequency of 4 Hz for each marker (Winter, 2005).

Joint angles were calculated as illustrated in Figure 4, with the anatomical position defined as 0° for all joints. Angles were measured at three times during the forward approach. Touchdown was defined as the first frame to which the diver's feet landed on the springboard (Jones & Miller, 1996). Maximum depression was the frame with the greatest downward deflection of the 5th metatarsal. Takeoff was defined as the final frame prior to the feet leaving the springboard (Sanders & Gibson, 2000). The angle of lean was measured at takeoff and defined by the angle between the right hand horizontal and a line from the centre of gravity to head of the 5th metatarsal.

Centre of mass location throughout the dive was calculated using the methods of Winter (2005) using the digitised segment locations and individual subject inertial parameters determined from Yeadon (1990). Dive height was calculated from the difference between the height of the centre of gravity at takeoff and at peak height. The height and vertical velocity of the subjects' centre of mass were recorded at touchdown and takeoff. The external work performed by the diver during the period of takeoff was calculated as the change in kinetic energy and potential energy between touchdown and takeoff (Sanders & Wilson, 1988). Mass was removed from the work equation to normalise the results between subjects (Eq. 1).

$$W = (g \cdot H_{TO} - g \cdot H_{TD}) + \left(\frac{1}{2} V_{TO}^2 - \frac{1}{2} V_{TD}^2 \right). \quad (1)$$

W = External Work ;

g = Acceleration due to gravity ;

H = Height of centre of mass at Takeoff (TO) and Touchdown (TD); V = Velocity of centre of mass.

A stepwise analysis of linear regression was used to identify factors that were most highly correlated with peak dive height. Lower limb joint angles at touchdown, maximum depression and takeoff; centre of mass vertical displacement at touchdown and takeoff; centre of mass velocity at touchdown; and external work done between touchdown and takeoff were all input as independent variables to predict peak dive height as the dependant variable. The regression analysis added independent variables sequentially, in order of correlation with the dependent variable, with $p < 0.05$ adopted as the criteria for entering variables. Variables were removed from the regression if their p values subsequently rose above 0.1 with the addition of further variables. After analysis of the results, a second regression was performed adding subject height as an additional independent variable.

3 Results

Regression analysis demonstrated that 73.8% of variance in peak dive height could be explained through the

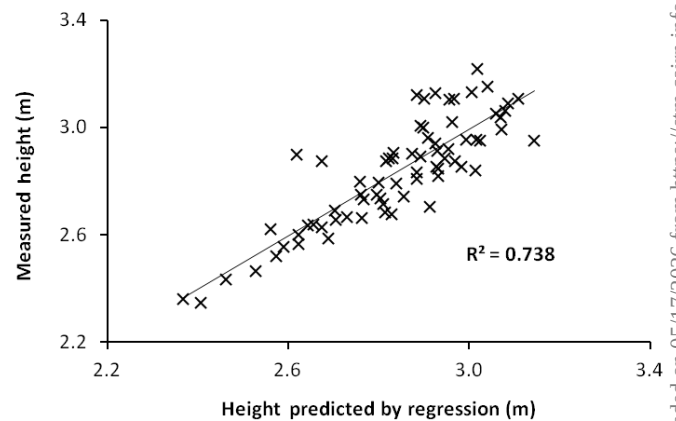


Fig. 5. Comparison between measured dive height and that predicted using linear regression.

addition of five independent variables: vertical velocity at touchdown (V_{TD}), vertical displacement at touchdown ($Height_{TD}$), angle of lean at takeoff ($Lean$), ankle angle at touchdown ($\theta_{AnkleTD}$) and external work ($Work$) (Fig. 5; Tab. 2; Eq. (2)). Each variable was entered stepwise into the regression equation in the sequence indicated by Table 2.

$$DiveHeight = -0.405 - 0.342 V_{TD} + 1.034 Height_{TD} - 0.018 Lean + 0.005 \theta_{AnkleTD} + 0.046 Work. \quad (2)$$

When the regression was repeated with subject height as an included variable, vertical displacement at touchdown and external work were no longer significant variables (Tab. 3), however the other independent variables remained with similar coefficients of regression (Eq. (3)).

$$Dive Height = 0.111 - 0.341 V_{TD} + 0.011 SubjectHeight - 0.022 Lean + 0.004 \theta_{AnkleTD}. \quad (3)$$

4 Discussion

The strongest predictor of overall dive height was vertical velocity at touchdown ($\Delta R^2 = 0.434$). This value is similar to the 50% contribution reported by Sanders and Wilson (1988) and comes as no surprise because, with more velocity at touchdown, there is more kinetic energy available to store as elastic energy in the board during compression. This finding emphasises the importance of the hurdle flight to overall dive performance.

Vertical displacement of the centre of mass had the second strongest correlation with dive height. While it is tempting to suggest that divers should endeavour to adopt a higher position at touchdown, subsequent analysis did not support this contention. Given the age of the divers within this sample (13.25 ± 1.04 years), there was significant physical growth between the youngest and oldest subjects. Therefore, the height at takeoff is more

Table 2. Regression analysis excluding subject height as an independent variable.

Variable	Coefficient	ΔR^2	Significance
1. Constant	-0.405		0.230
2. Vertical velocity at touchdown	-0.342	0.434	0.000
3. Vertical displacement at touchdown	1.034	0.021	0.000
4. Angle of lean at takeoff	-0.018	0.105	0.000
5. Ankle angle at touchdown	0.005	0.023	0.004
6. External work	0.046	0.024	0.003

ΔR^2 is the change in correlation for the overall regression as each stepwise variable is entered.

Table 3. Regression analysis including subject height as an independent variable.

Variable	Coefficient	ΔR^2	Significance
1. Constant	0.111		0.683
2. Vertical velocity at touchdown	-0.341	0.434	0.000
3. Subject height	0.011	0.162	0.000
4. Angle of lean at takeoff	-0.022	0.121	0.000
5. Ankle angle at touchdown	0.004	0.021	0.023

ΔR^2 is the change in correlation for the overall regression as each stepwise variable is entered.

likely to be representative of the strength of the subjects than of a deliberate adoption of a high landing position. Re-running the regression statistics using subject height as an independent variable eliminated touchdown height as a significant predictor of dive height. This also matches previous recommendations (Miller & Monro, 1984) for adopting a knees-flexed position at impact (Fig. 6) in order that the legs can extend during board compression, storing further elastic energy within the board.

The angle of lean had a negative correlation with dive height, indicating that a more upright position was associated with higher flight characteristics. This finding has previously been reported by Sanders and Wilson (1988). Adopting a greater lean at takeoff results in increased rotational momentum during the dive (Miller & Springings, 2001) and presumably also a greater horizontal velocity off the board; both of which would leave less kinetic energy available for vertical velocity at takeoff. The position at takeoff for a somersault therefore becomes a compromise between the need to lean forward to develop angular momentum and the requirement to be more vertically aligned for dive height.

Sanders and Wilson (1988) have previously reported a significant correlation between hip angle at takeoff and

**Fig. 6.** Diver shown at the point of impact with the board.

dive height. No such relationship was found within the present study and the difference is likely to result from different definitions of hip angle between the two studies. The present study used markers on the hip centre and iliac crest to define a pelvic angle; with hip angle being the difference between thigh and pelvic angle (Fig. 4). By contrast, Sanders and Wilson (1988) used markers on the hip and shoulder to define a trunk angle; with hip angle being the difference between trunk and thigh angles. Their finding of increased hip angle, therefore, is likely to have resulted from increased flexion of the lumbar vertebrae associated with positions of greater forward lean, rather than actual flexion of the hip *per se*.

Ankle angle at touchdown was a small but statistically significant predictor of dive height. The positive correlation indicates that increased dorsiflexion at touchdown was associated with greater dive height. A possible mechanism for this may be that eccentric work across the ankle absorbs energy if the foot is not flat on the board at impact. Alternately, a more dorsiflexed position may allow a greater range through which the ankle can extend, adding energy during the takeoff.

External work done during takeoff was calculated from the divers' change in mechanical energy between touchdown and takeoff. Although the initial regression identified external work as a significant independent variable, the contribution of this variable to total dive height was relatively small. For example, using the average work done on the board (3.3 J/kg) and the regression coefficient of 0.046, external work contributed 0.152 m, only 5.4% of total dive height. By contrast, the average contribution of vertical velocity at impact was $-0.342 \times -0.56 = 1.218$ m; approximately 43% of total dive height. After adding subject height as an independent variable to the regression analysis, external work was no longer identified as a significant predictor. Therefore, even with mass removed from the work

calculations, this suggests simply that the taller subjects were able to perform more work on the board and not that there are technique implications for individual divers.

Further research is required to fully understand the implications of the present study. Firstly, this study has emphasised the importance of generating a high hurdle flight in order to impact the board with as much velocity as possible for the final take-off. A similar study looking at characteristics of the hurdle flight may be beneficial in understanding how the height of the hurdle can be maximised. A second recommendation is to look more closely at the velocity of the board and how it interacts with the diver. Catching the board at its maximum downwards velocity has been recommended as a means to minimise energy loss during initial impact with the board. Board velocity data was not available within the present study, but would have made a valuable addition to our understanding. Knowing the board velocity at impact would allow us to assess how well the board was “caught” by the diver. Further, knowing the board’s velocity at touchdown and takeoff, together with some estimate of the effective mass of the board, may have allowed calculation of the amount of energy gained or lost from changing the board’s velocity during the takeoff. A third recommendation is to increase the number of trials analysed for each subject in order to look specifically at within-subject variability. Factors that correlate with performance across a group of subjects may not be important when considering the performance of an individual (Salter, Sinclair, & Portus, 2007). To do this, more automated means of data collection would be required given the 90–120 min per trial spent digitising the current data.

5 Conclusions

- The variable most strongly correlated with dive height was vertical velocity at touchdown. This emphasises the need to analyse actions of the hurdle in order to better understand jump performance.

- Vertical displacement at touchdown, although initially identified as a predictor, was related to the height of participants. Once height had been included as an additional independent variable, vertical displacement no longer had a significant input. Therefore, there are no technique recommendations for individual divers’ landing height from this study.

- Ankle angle at touchdown was correlated with height, with more dorsiflexed positions being associated with greater heights. It is possible that greater plantarflexion at touchdown may increase eccentric absorption of energy across the ankle during the initial time of contact with the board, thus reducing jump height.

- External work done on the board was a significant predictor of jump height but its contribution was relatively small; accounting for only about 6% of total jump height. As with displacement at touchdown, when subject height

was added as an independent variable, work was eliminated from the regression equation.

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